THE CURE CENTER
FOR ULTRASOUND RESEARCH & EDUCATION

INTERCRANIAL CEREBROVASCULAR ULTRASOUND SYLLABUS

COURSE CODE: DMS 402

INTRACRANIAL CEREBROVASCULAR ULTRASOUND

Instructor:
Telephone:
E-Mail:
Office Hours:
Mon, Wed, Fri: 12pm to 1pm & 5pm to 6pm
Tues: 9am – 5pm
Thurs: 11am – 5pm

Start Date:

Completion Date:

Days and Times:
Monday, Wednesday and Friday

Course Hours:
Morning Session:
8:00 am -10:00am &
10:10am – 12:00 pm
Afternoon Sessions:
1:00pm – 3:00pm &
3:10 pm – 5:00 pm

Class Location:
Classroom #:
Lab#:

Prerequisites:
BIO 100 Introduction to A&P
PAT 100 Patient Care
LAW 100 Medical Law & Ethics

Course Contact Hours: 60
Lecture: 24
Laboratory: 36
Course Delivery: Residential
Pre-requisite: None

Americans with Disabilities Act Guidelines:

Requests for accommodations should be submitted to the campus president. Faculty are not authorized to grant accommodations.

COURSE DESCRIPTION

This is an introductory level course directed towards developing the principles, techniques, knowledge and skills of each student necessary to function as a qualified diagnostic medical sonographer in the everyday practice and application of vascular sonography. It introduces the diagnostic foundations of vascular sonography, terminology, abbreviations, scan plane orientations, directional nomenclature, vascular hemodynamic, anatomical relationships, and normal variance and pathologic finding.

Emphasis will be placed on the academic and laboratory applications within this highly specialized area of sonography including basic patient care principles of vascular patient, professionalism, interpretation of normal and abnormal sonographic patterns and features, pathology, associated clinical laboratory tests and procedures and sessions on basic scanning techniques and vascular protocols.

Instructional Materials

1. Textbook
Title: Textbook of Diagnostic Sonography, 8th Edition (2 Volume Set)
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323353755

2. Workbook
Title: Workbook for Diagnostic Sonography, 8th Edition
Author(s): Sandra L. Hagen-Ansert, MS, RDMS, RDCS, FASE, FSDMS
Publisher: Mosby; 8th edition (May 3, 2017)
ISBN: 978-0323441834

3. Textbook
Title: The Vascular System, 2nd Edition
Author(s): Ann Marie Kupinski, PhD,RVT, RDMS, FSVU
Publisher: Wolters Kluwer Copyright © 2018
ISBN: 978-1496380593

4. Textbook
Title: Pocket Protocols for Sonography Scanning, 4th Edition
Author(s): Betty Bates-Tempkin, BA, RT®, RDMS
Publisher: Saunders; 4th edition (December 27, 2014)
ISBN: 978-1455773220

Equipment/Technology

- LCD Projector
- Laptop Computer
- Mindray Ultrasound Machine

Material

- Electronic and Printed Worksheets
- Textbooks
- Visual Aids: DVD Videos, Case Study Reviews, Demonstrations and Hands on Clinical Laboratory Practice.

During clinical laboratory sessions various techniques such as role playing and scanning exercises will be used.

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Grade Point Value

Grade	Percentile Range	Points
A+	97-100	4.00
A	93-96.99	3.85
A-	90. -92.99	3.67
B+	87-89.99	3.33
B	83-86.99	3.00
B-	80-82.99	2.67
C+	77-79.99	2.33
C	73-76.99	2.00
C-	70-72.99	1.67
D+	67-69.99	1.33
D	63-66.99	1.00
D-	60-62.99	.67
F	Below 60-00	0

Methods of Evaluation

Exams / Quizzes/ Midterm/ Final	40%
Projects/ Attendance Participation/ Homework Assignments	20%
Scan-out	40%
	100%

Teaching Strategies

A variety of teaching strategies may be utilized in this course, including but not limited to: lecture, discussion, written classroom exercises, visual aids, videos, case study reviews, group and/or individual projects, quizzes, exams, reading assignments, writing assignments, interactive exercises demonstrations and hands, internet research and hands on clinical laboratory practice.

Examination Standards

Students are expected to be honest in all academic work. Cheating of any kind is not tolerated. Violations of this principle will result in a grade of zero in the course and additional disciplinary action including expulsion as determined by CURE.

COURSE OBJECTIVES

On completion of this course students should be able to:

- Distinguish normal from abnormal intracranial arterial anatomy.
- Understand the physiology and hemodynamics within the circle of Willis.
- Outline the technical aspects and proper instrumentation. Control settings used to perform a high-quality transcranial Doppler imaging examination.
- Describe the characteristics of the spectral Doppler waveforms obtained from arteries evaluated during a transcranial Doppler examination.
- List the clinical applications of transcranial Doppler imaging and describe expected findings and diagnostic criteria associated with each.
- Discuss common errors associated with performing and interpreting a transcranial Doppler imaging examination.
- List the different clinical applications of vascular imaging and describe expected findings and diagnostic criteria associated with each.
- Discuss common errors associated with performing and interpreting vascular imaging.
- Explain how artifacts can be handled to avoid the pitfalls and misdiagnoses that they can cause.
- Comply with ALARA's "Policy on Ultrasound Safety" and the use of the equipment safely.
- Label and save the images of the normal or pathologic vascular appearance pattern.
- Discuss differential diagnoses and case presentations.
- Successfully perform different vascular scanning protocols, independently.

- Recognize and discuss ultrasound results based on class discussions, video, lecture and PowerPoints.
- Pass the didactic course with a minimum passing grade of 73.

Attendance Policy

Routine, regular class attendance is a requirement for this course.
Promptness and dependability are the soft skills that garner special attention

Attendance Rubric:

Up to 25% will be deducted for each hour missed or for any part thereof (5% for each 10 minutes missed).
Students are required to meet their professor to discuss matters of tardiness or absence.

Students are required to notify their instructor via E mail or voicemail to communicate an absence or lateness from class

A student absent from all of his or her classes for a period of 14 consecutive calendar days during which classes or examinations are in session will be regarded as having withdrawn from the program and will receive a grade of W for each course in which he or she is enrolled.

HOMEWORK

An essential part of every program of study, homework enables students to master course learning objectives and leads toward the achievement of the overall program objectives. Homework refers to self-directed learning activities such as reading assignments, research activities and projects, quiz/exam preparation, knowledge checks and self-assessments, etc.

Late Homework/Make-Up Policy

Late homework will be accepted under the following guidelines:

1. It must be turned in within two business days (Monday through Thursday) after the due date.
2. However, homework turned in after the due date will be penalized 10% for each day it is late up to two calendar days after the due date.
3. Homework submitted after three business days after the due date will earn a zero.

Exams and quizzes - Make up of a quiz or an exam will be allowed under the following extenuating circumstances only:

- Documented illness
- Military duty
- Death of a family member
- Court appearance or jury duty

Documentation of the above circumstances must be provided to the Program Director. ***The ability to make-up a quiz or exam must be approved by the Program Director.***

All assignments should be submitted directly to the instructor.

Classroom Policy

Our goal is to build a respectful learning and work environment that allows for positive communication and teamwork. To support your development, all students must abide by academic policies related to attendance, behavior and professional conduct as published in the current institutional catalog. Special attention is granted to the following items:

- There is to be no eating or drinking in any lab/computer class.
- There is to be no eating in ANY class.
- Cellular phones may only be used during class if/when advised by the instructor and required for a learning activity.
- All students must be in full uniform which includes white or black shoes at **ALL TIMES.**
- **There are no rescheduled exams *unless specified by the professor* with advance notification.**
- The determination to accept late homework assignments is made only by the instructor and occurs only in the event of extraordinary circumstances. If the professor does allow for an extension on assignments, there will be an automatic daily reduction by 10 points, each day beyond the original due date.
- All classes are expected to start and end at a predetermined start and end time.

Accommodations for Students with Disabilities

If you have a special learning need, please contact the Program Director. The Program Director and the Campus President will work with the student to identify reasonable accommodations or adjustments necessary to enable him or her to fully participate in the educational processes.

Professional Responsibility

Students are expected to aspire to the same high level of ethics and professionalism that is demanded of registered Ultrasound students. This requires students to conduct themselves as professionals and to fully engage in the learning process by preparing for and participating in all classroom, laboratory and clinical activities.

Attendance is taken each hour of class by the instructor and turned over to the Registrar before the end of the class day. Early departures, tardiest, and class cuts will be recorded in quarter-hour increments and a proportional grade reduction will also be applied. A period of less than 15 minutes will be counted as a quarter-hour of absence. Attendance records are maintained by the Registrar as part of the student's permanent academic record.

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The course syllabus is a general plan for the course; deviations may be necessary and will be announced.

COURSE OUTLINE
Below is the course schedule which identifies lecture topics, assignments and other course activities and the homework required for completion each week.

Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week One Objectives: • Distinguish normal from abnormal intracranial arterial anatomy. • Understand the physiology and hemodynamics within the circle of Willis. • Outline the technical aspects and proper instrumentation. Control settings used to perform a high-quality transcranial Doppler imaging examination.		
DAY 1: **Intracranial Arterial Anatomy** Internal Carotid Artery Ophthalmic Artery Vertebral Arteries Basilar Artery	Power Point Presentation/Lecture	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 1 and Exercise 2 Pages 299
DAY 2: **Intracranial Arterial Anatomy** Circle of Willis Middle Cerebral Artery Anterior Cerebral Artery Anterior Communicating Artery	Power Point Presentation/Lecture	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 3 Pages 300
DAY 3. **Intracranial Arterial Anatomy** Posterior Cerebral Arteries Posterior Communicating Artery Intracranial Arterial Physiology	Power Point Presentation/Lecture	**Read** Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 4 Page 301

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Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week TWO Objectives: • Describe the characteristics of the spectral Doppler waveforms obtained from arteries evaluated during a transcranial Doppler examination. • List the clinical applications of transcranial Doppler imaging and describe expected findings and diagnostic criteria associated with each. • Discuss common errors associated with performing and interpreting a transcranial Doppler imaging examination.		
DAY 4. Intracranial Arterial Physiology Technical Aspects of Transcranial Doppler Imaging Instrumentation Technique	Power Point Presentation/ Lecture	Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 5 page 301 Page 45-47
DAY 5. Technical Aspects of Transcranial Doppler Imaging Instrumentation Technique Interpretation of Transcranial Doppler Imaging	Power Point Presentation/ Lecture	Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 6 Page 302
DAY 6. Physiologic Factors Normal Intracranial Velocities Clinical Applications Vasospasm Diagnosis of Intracranial Disease Intraoperative Use of Transcranial Doppler Technology	Power Point Presentation/ Lecture	Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 7 Pages 302

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Lecture Topic	In-Class Activity/Lab Assignment	Homework/Out of Class
Week Three Objectives: • Clinical Applications • Review/Scan-Out/Final		
DAY 7. Clinical Applications Vasospasm Diagnosis of Intracranial Disease Intraoperative Use of Transcranial Doppler Technology Advantages and Limitations of Transcranial Doppler Imaging Diagnostic Pitfalls	Power Point Presentation/ Lecture REVIEW	Textbook: Diagnostic Sonography 8ed Sandra Hagen Chapter 38: Intracranial Cerebrovascular Evaluation Pages. 983-1005 Chapter 38 Homework: Workbook Assignments. Exercise 8 Page 303
DAY 8. SCANOUT/FINAL	FINAL EXAM	SCANOUT/FINAL